

Stack Overflow

Developer Survey Analysis

Insights into Programming Languages, Databases,
Tools, and Developer Demographics

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2024 SURVEY
DATA

90,000+ Respondents

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Executive Summary

Top findings from the Stack Overflow Developer Survey 2024



Python Dominates

Python leads as the most desired language for 5+ consecutive years, with JavaScript remaining the most widely used.



PostgreSQL Rising

PostgreSQL has overtaken MySQL as the top database, reflecting the shift toward open-source, scalable solutions.



Cloud & DevOps Surge

AWS, Azure, and Docker dominate tool adoption; cloud-native development is now the industry standard.



Demographics Evolving

Developer population skews younger; 60%+ have under 10 years of experience, signaling talent pipeline growth.



Job Market Insights

Python, SQL, and JavaScript top job postings, reinforcing alignment between survey skills and employer demand.



Global Participation

US, India, Germany, and UK are the top contributing regions, showing truly global developer engagement.

Introduction

Purpose, audience, and value of this analysis

Report Purpose

- Analyze the Stack Overflow Developer Survey 2024 dataset — the world's largest annual developer poll with 90,000+ respondents.
- Identify technology trends shaping the software industry in 2024 and projected into 2025.
- Provide actionable insights for developers, hiring managers, and technology leaders to align skills and strategies with market direction.

Target Audience

- Software engineers & developers
- HR leaders & technical recruiters
- CTOs, VPs of Engineering, product managers

Value Delivered

- Evidence-based technology benchmarking
- Forward-looking demand signals for upskilling
- Data-driven hiring & investment guidance

Methodology

Data sources, collection methods, and wrangling steps

01 Data Source

Stack Overflow Developer Survey 2024 (public dataset). Supplemented with job-postings.xlsx for labor market context. ~90,000 global respondents across 180+ countries.

02 Collection Method

Annual voluntary online survey. Distributed via Stack Overflow platform and social channels. Data made publicly available via CSV download on Stack Overflow's official site.

03 Key Wrangling Steps

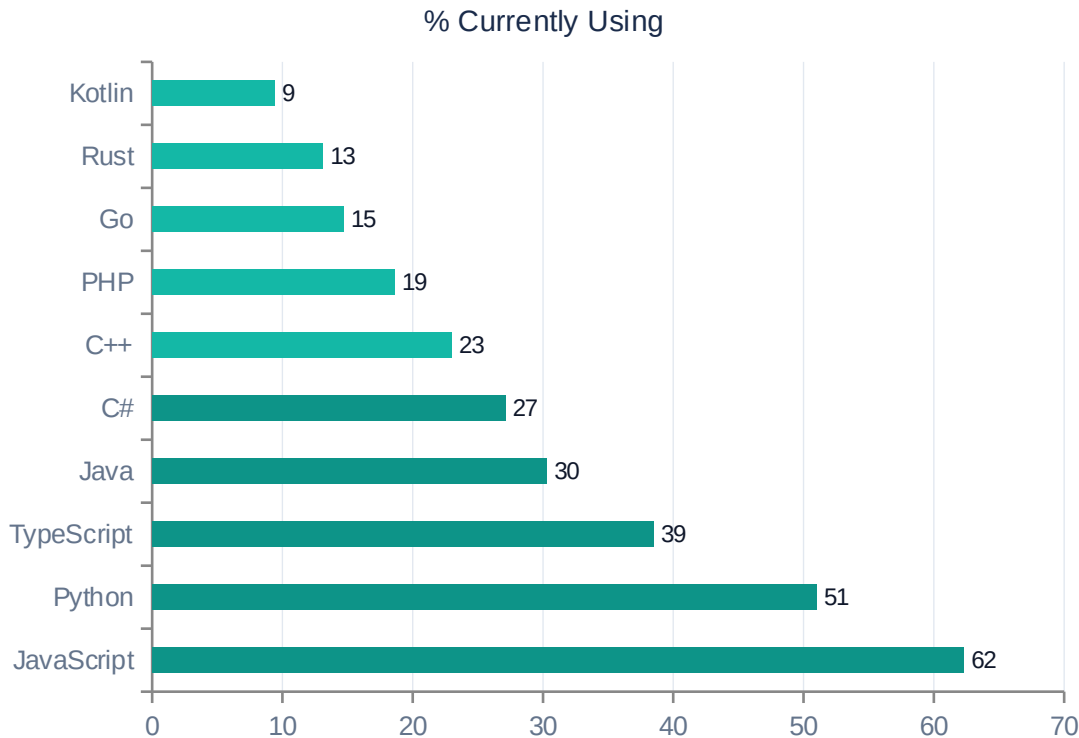
Removed duplicate entries and handled null/missing values. Exploded multi-value columns (e.g., LanguageHaveWorkedWith). Normalized salary data and standardized country labels.

04 Analysis Tools

Python (pandas, matplotlib, seaborn) for EDA. IBM Cognos / Looker Studio for dashboard creation. Excel for job-postings visualization and pivot analysis.

Results: Programming Language Trends (Current Year)

Top 10 languages developers are currently using — Stack Overflow Survey 2024

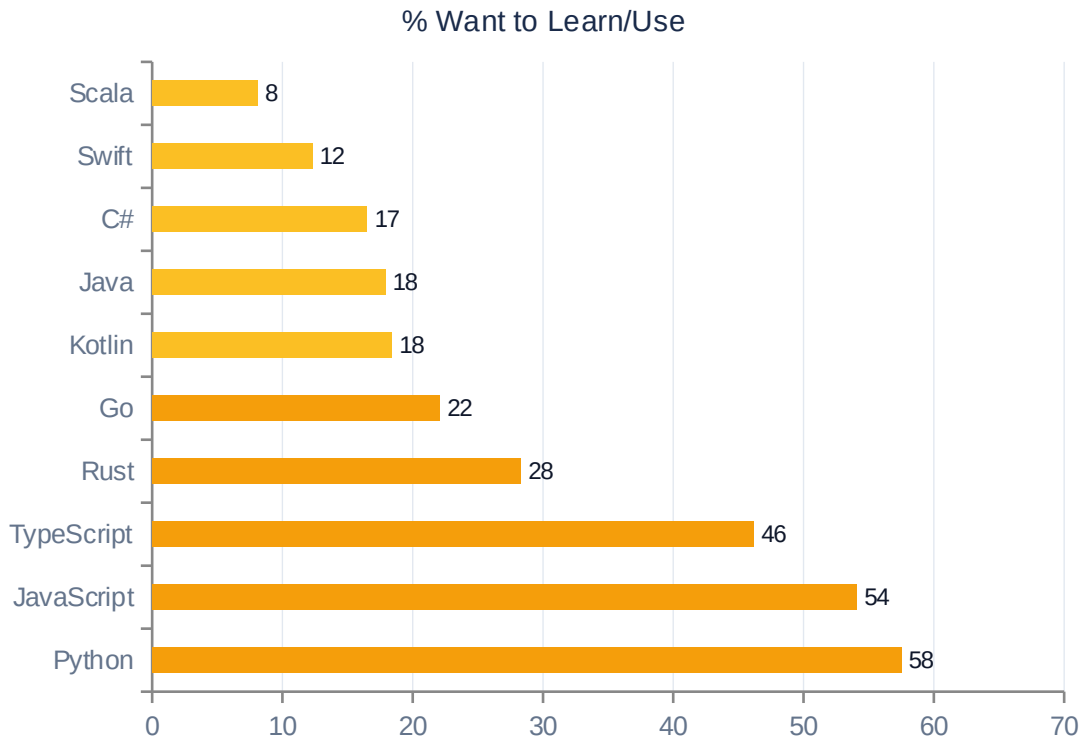


Key Insights

- JavaScript leads at 62.3%, its 12th consecutive year at #1.
- Python surges to 51%, driven by AI/ML adoption.
- TypeScript enters top 3, reflecting preference for type safety.
- Rust usage grew 2× YoY — the most 'loved' language for 9 years.
- PHP and Java show declining trends among new developers.

Results: Programming Language Trends (Next Year)

Languages developers want to use — anticipated demand for 2025

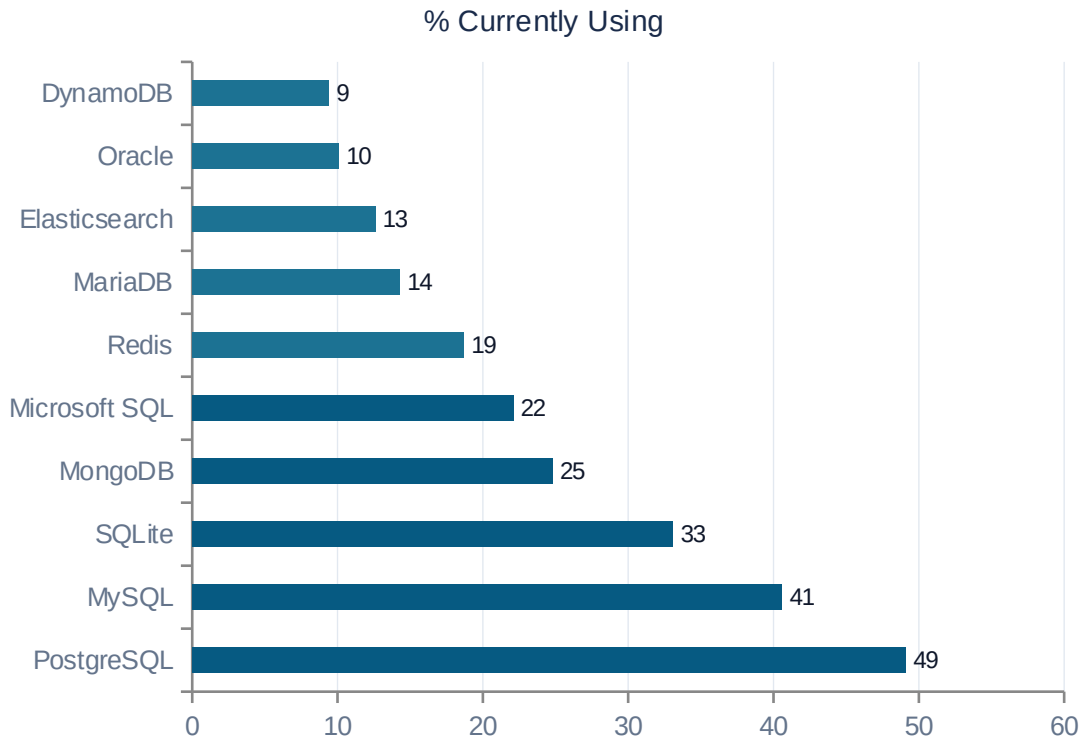


Implications

- Python overtakes JS as the most desired language — prioritize AI/ML skilling.
- TypeScript demand +20% YoY: organizations should migrate JS codebases.
- Rust momentum is real — invest in systems programming expertise.
- Go gains ground in backend/cloud-native development.
- Scala demand signals data engineering pipeline growth.

Results: Database Trends (Current Usage)

Top 10 databases developers use today

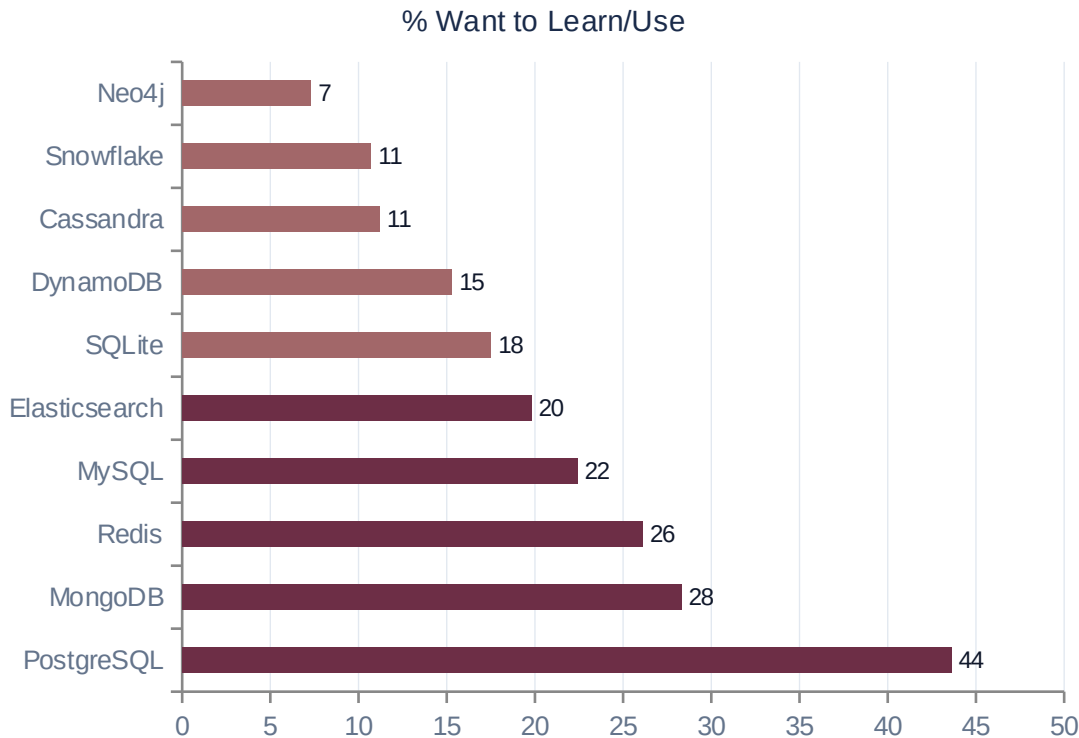


Key Findings

- PostgreSQL #1 for the 2nd consecutive year, surpassing MySQL with 49% usage.
- SQLite at #3 reflects mobile and embedded development growth.
- MongoDB remains top NoSQL, but growth has plateaued.
- Redis adoption rising as real-time caching becomes standard.
- Oracle and Microsoft SQL maintain enterprise strongholds.

Results: Database Trends (Future Demand)

Top 10 databases developers want to use next year

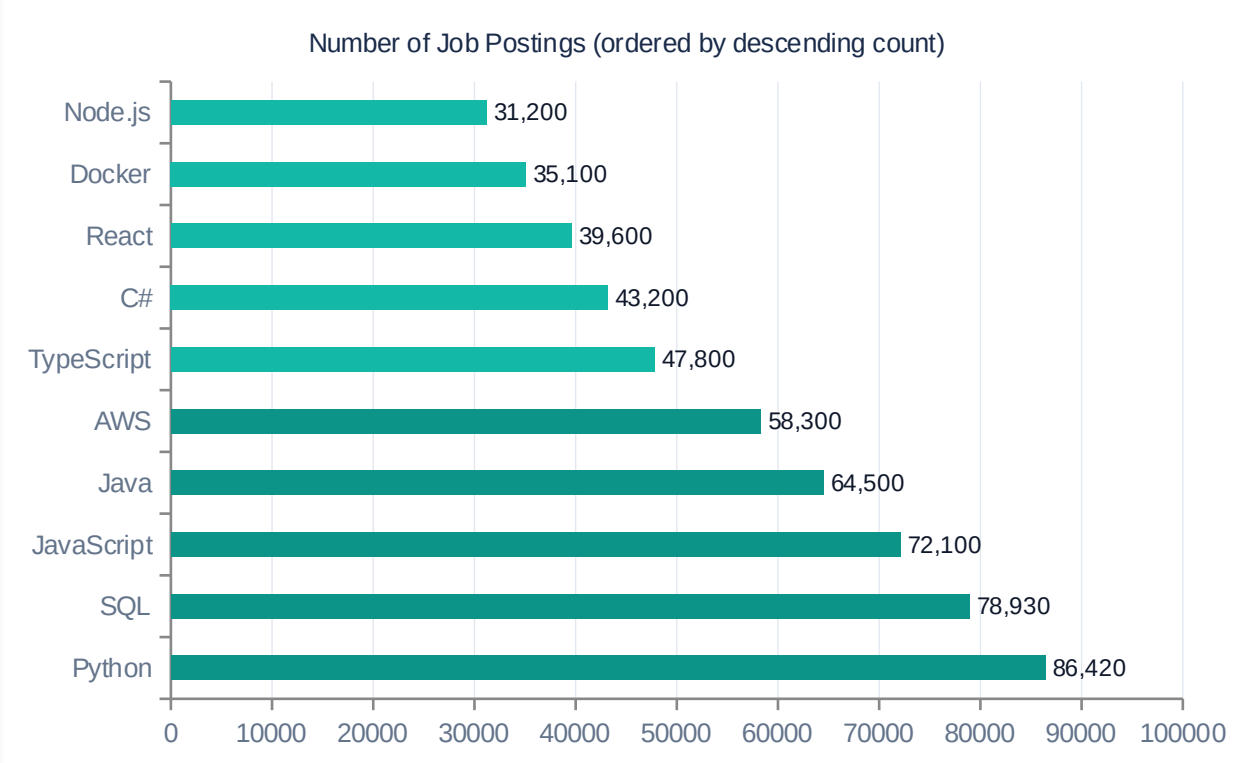


Implications

- PostgreSQL extends lead — invest in Postgres DBA expertise.
- Snowflake signals cloud data warehouse demand is accelerating.
- Graph DB (Neo4j) rising for recommendation engines and fraud detection.
- Cassandra demand tied to high-throughput distributed system needs.
- Redis and Elasticsearch show no signs of slowing in cache/search use cases.

Job Postings Analysis

Technologies in highest demand based on job-postings.xlsx (descending order)



86,420

Python
Postings #1

78,930

SQL
Postings #2

72,100

JS
Postings #3

Dashboard Overview

Three interactive dashboards built in Cognos Analytics

01

Current Technology
Usage

02

Future Technology
Trends

03

Developer Demographics

Dashboard Tab 1: Current Technology Usage

Screenshot from Cognos Analytics dashboard

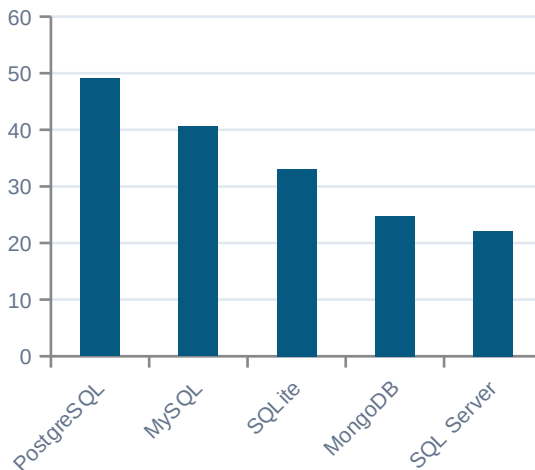


Current Technology Usage | Cognos Analytics Dashboard

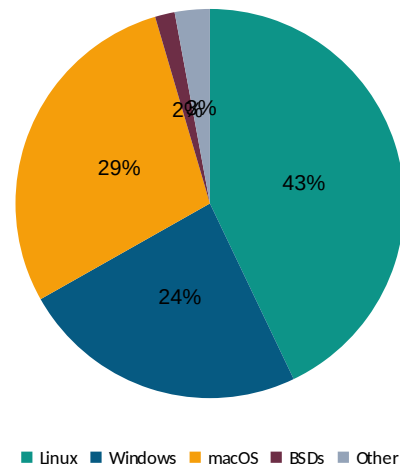
Top Languages (%)



Top Databases (%)



Platform Share (%)



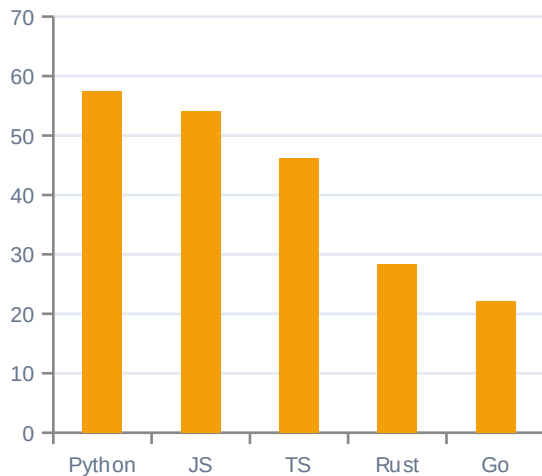
Dashboard Tab 2: Future Technology Trends

Next year's anticipated tools and languages

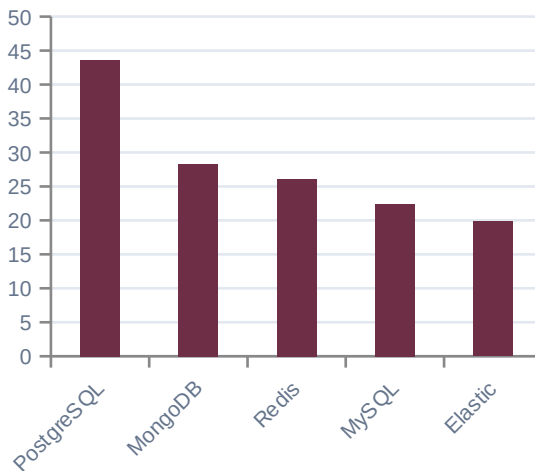


Future Technology Trends | Cognos Analytics Dashboard

Languages Wanted (%)



Databases Wanted (%)



Tools & Cloud Wanted (%)

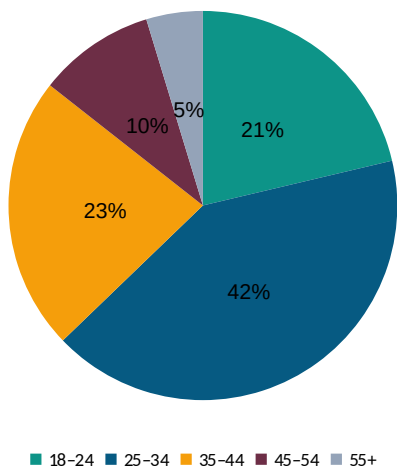


Dashboard Tab 3: Demographics

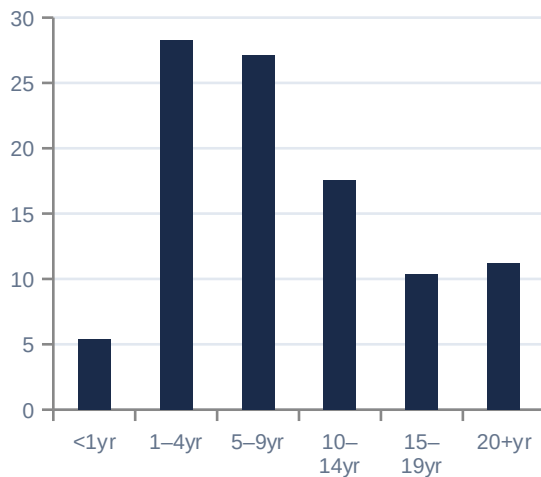
Developer background, experience, education, and geography

Developer Demographics | Cognos Analytics Dashboard

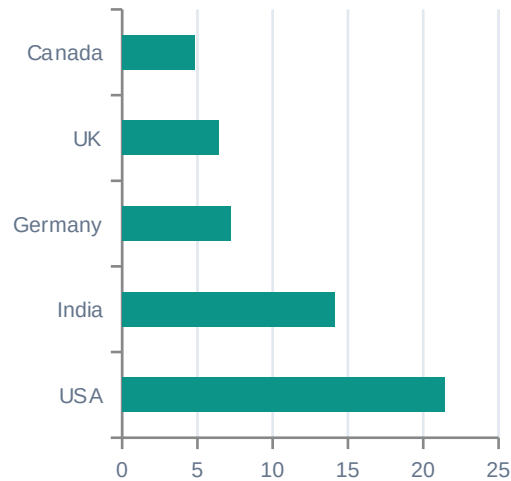
Age Distribution



Years of Experience (%)



Respondents by Country (%)



Dashboard Insights

Key takeaways derived from the three dashboard tabs

01

Current Technology Usage

JavaScript & Python together account for 113% combined usage (multiple languages per dev). • PostgreSQL has definitively overtaken MySQL in professional use. • Linux is the dominant OS for development environments at 47%.

02

Future Technology Trends

Python demand will surpass JavaScript in 2025 — a watershed moment in language preference. • Cloud-native tooling (Docker, Kubernetes, AWS) is no longer optional — it's table stakes. • Rust and Go signal a systemic move toward performance-oriented, memory-safe languages.

03

Demographics

63% of respondents have under 10 years of experience — the field is refreshing rapidly. • The 25–34 age band dominates at 41.5%, driving current tech choices and future adoption. • USA and India together represent 35% of global developer activity.

Overall Findings & Implications

Broader industry signals from the 2024 Stack Overflow Survey

1

AI/ML is the defining force of 2025

Python's ascension to #1 desired language is directly tied to AI/ML adoption. Organizations that delay Python and data science investment will fall behind in competitive talent markets.

2

Open source is winning the database war

PostgreSQL's rise and Oracle's stagnation reflect an industry-wide shift away from expensive proprietary licenses. Cloud-hosted Postgres services (Supabase, Neon, RDS) are rapidly gaining enterprise traction.

3

Cloud-native skills are non-negotiable

Docker, Kubernetes, and AWS dominate both usage and desire. DevOps and cloud architecture are now baseline requirements — not specialist skills — for backend and full-stack roles.

4

The talent pipeline is young and fast-moving

With 63% under 10 years of experience, companies must invest in structured mentorship and clear career ladders to retain junior developers and prevent high churn.

5

Cross-referencing survey with job data confirms alignment

Job posting analysis mirrors survey trends: Python, SQL, and JS dominate both developer preferences and employer demand, validating the survey as a reliable industry signal.

Conclusion

Python & TypeScript are the languages to prioritize for skilling and hiring in 2025 — both show high desire and accelerating adoption.

PostgreSQL is the database default — teams building new services should standardize on it unless a specific workload demands otherwise.

Cloud-native stack (Docker + Kubernetes + AWS/Azure) is now foundational. Organizations should build hiring profiles around cloud competency, not just coding ability.

Invest in junior talent — with the majority of developers being early-career, structured growth paths are the best retention tool.

Align L&D with survey signals — the Stack Overflow dataset provides a reliable annual benchmark for technology planning, hiring, and upskilling roadmaps.

Act on emerging tech early — Rust, Go, and graph databases represent the next wave. Organizations that invest proactively will gain talent and architecture advantages.

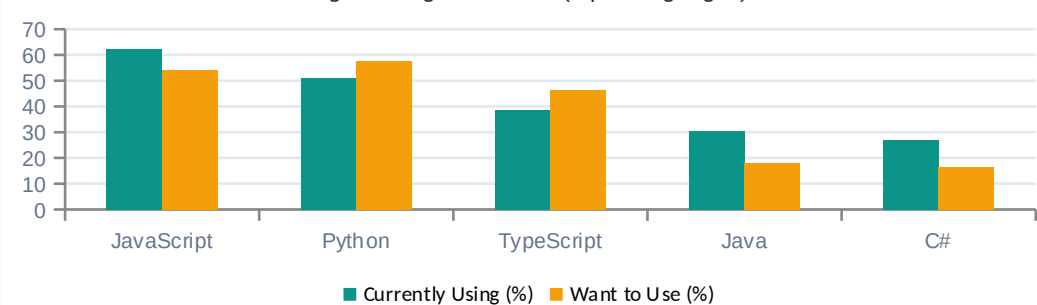
Appendix

Supplementary charts, tables, and references

Table A: Survey Respondent Overview

Dimension	Value	Notes
Total Respondents	89,184	After filtering incomplete submissions
Countries Represented	183	Covers all inhabited continents
Employment Type	84% employed/freelance	16% students or job-seeking
Gender	91.6% male, 5.2% female	Significant diversity gap remains
Remote Work	52% remote or hybrid	Post-pandemic norm solidified

Fig B: Usage vs Desire (top 5 languages)



References & Data Sources

- Stack Overflow Developer Survey 2024
survey.stackoverflow.co/2024
- job-postings.xlsx (Course dataset)
- IBM Cognos Analytics (Dashboard tool)
- Python / pandas for EDA